

Shruti Debnath

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

EDUCATION

University of Idaho | *Recipient of the Invitation to Idaho Scholarship*

Idaho, USA

Bachelor of Science in Computer Science | AI/ML Focus

December 2026

GPA: 3.5/4.0 | Dean's List (4 semesters)

Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering, Machine Learning

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C#, C, C++
- **Frameworks & Libraries:** React, Next.js, Node.js, Django, Tailwind CSS, Framer Motion, REST APIs
- **Databases:** PostgreSQL, MySQL, SQLite
- **Software and Developments:** REST APIs, Object-Oriented Programming, Git, Unit Testing, API Testing, Debugging, Vercel, Render, Unity
- **AI-Assisted Development:** GitHub Copilot, ChatGPT, Claude, Cursor

PROJECTS

CATScan — Automated Security Testing Tool for C/C++

Python, Regex, LibClang, C/C++, HTML

- Engineered an automated security-testing pipeline for C/C++ programs using Python and libclang, combining static analysis with runtime validation to detect common software vulnerabilities.
- Implemented automated test generation for buffer overflow, integer overflow, divide-by-zero, blank-input, and format-string vulnerabilities, improving repeatability of defect detection.
- Automated compilation, execution, timeout handling, failure diagnosis, and HTML report generation to surface runtime errors and detected vulnerabilities.

Barter App

Flask, REST APIs, Python, SQLite, HTML, Git

- Designed and developed a full-stack barter platform using Flask, Python, SQLite, and REST APIs, including a normalized 6-table relational database for users, items, trades, partnerships, and transactions.
- Implemented 26+ REST API endpoints supporting authentication, item management, trade proposals, user profiles, partnerships, and administrative workflows.
- Designed a two-step transaction-verification workflow using split 16-digit codes to prevent unilateral confirmation and improve transaction integrity.

Diabetes Classification

Python, scikit-learn, Machine Learning

- Developed and evaluated supervised machine learning models using the Diabetes Health Indicators dataset to classify diabetes risk from structured health data.
- Implemented and compared Decision Tree, Logistic Regression, and Random Forest models using scikit-learn, analyzing model performance to identify the most effective approach.

Combat Game

Unity, C#, Git

- Led a 7-member development team in architecting a modular Unity/C# application, applying object-oriented design and dynamic binding across gameplay and UI systems.
- Designed 8+ reusable UI components with centralized configuration and Apply/Discard state management to improve maintainability and consistency.
- Developed 30+ unit and boundary tests plus 5 stress tests to validate gameplay and UI behavior while coordinating development through Git-based workflows.

WORK EXPERIENCE

SI-PASS Leader

Aug 2024 – Present

University of Idaho — College Success (Tutoring)

Idaho, USA

- Led weekly SI sessions for ~20 students, increasing participation and improving exam performance in Pre-Calculus.
- Contributed to ~10% improvement in average midterm scores among consistent attendees.
- Collaborated with course instructor to monitor discussions and resolve conceptual gaps in real time.

Lead Tutor

Oct 2025 – Present

University of Idaho — Math Assistance Center (MAC)

Idaho, USA

- Led bi-weekly strategy meetings with 14 tutors to align coursework updates and improve instructional consistency.
- Collaborated with supervisors to implement engagement-focused initiatives, increasing student attendance.
- Provided one-on-one and drop-in tutoring support to improve student performance in mathematics courses.
- Developed training materials aligned with CRLA certification standards.